MOSFET Small-Signal Characteristics

Amin Chegeni

2026-10-03

We will extract the small-signal characteristics of MOSFETs using freely available, open-source PDKs and CAD tools. The analysis is carried out using the SkyWater **sky130** PDK, **Xschem** as the schematic capture tool, and **Ngspice** as the circuit simulator.

All tools (Xschem, Ngspice, and sky130 PDK) must be installed properly on a Linux system. In this overview, Ubuntu running under WSL is used. The installation procedure is discussed in a separate article.

The reader is assumed to have a basic understanding of MOSFET device physics and the fundamentals of MOSFET operation. Figure 1 illustrates the structure and symbols of NMOS and PMOS devices. In integrated-circuit design, a MOSFET has four terminals: gate (G), source (S), drain (D), and bulk (B). Modern CMOS technologies typically employ a p-type substrate; therefore, the bulk of all NMOS devices (except native devices) is tied to the substrate, i.e., GND or VSS. In contrast, PMOS devices are fabricated in isolated wells, allowing their bulks to be biased independently.

Operation regions:

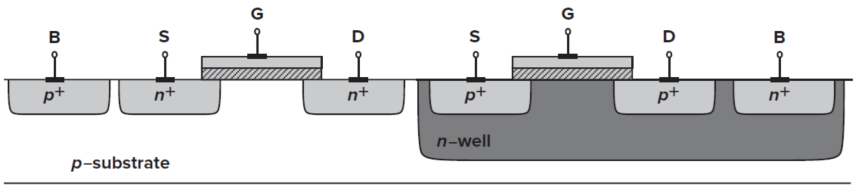
A MOSFET behaves as a switch (ON or OFF) in digital design, but in analog design it operates in one of three regions:

- Cut-off (off): $V_{GS} < V_{TH} \quad and \quad V_{GD} < 0$
- Saturation: $V_{GS} > V_{TH} \quad and \quad V_{GD} < 0$
- Triode (on): $V_{GS} > V_{TH} \quad and \quad V_{GD} > 0$

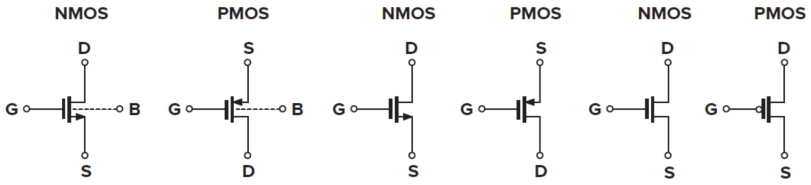
Note that the transition between these regions is smooth, especially near the threshold voltage.

Voltage–Current Characteristics:

To explain MOSFET operation, we review the V–I equations of an NMOS transistor. PMOS equations follow the same form.



(a)



(b)

Figure 1: (a) structure of NMOS and PMOS, (b) symbols (from [1])

Triode region (ON):

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH})\, V_{DS} - \frac{1}{2} V_{DS}^2 \right] \tag{1}$$

For $V_{DS} << 2(V_{GS} - V_{TH})$:

$$I_D \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})\, V_{DS} \tag{2}$$

Saturation region:

Including channel length modulation, the mosfet current is given by:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda\, V_{DS}) \tag{3}$$

The parameter λ is the channel-length modulation coefficient, which decreases with increasing channel length.

$$\lambda \propto \frac{1}{L} \tag{4}$$

Second-order effects:

Body effect:

Increasing the source-to-bulk voltage raises the threshold voltage:

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{2\Phi_F + V_{SB}} - \sqrt{|2\Phi_F|} \right) \tag{5}$$

$$\gamma = \frac{\sqrt{2q\epsilon_{si}N_{sub}}}{C_{ox}} \approx 0.3 \; to \; 0.4 \; \left(V^{\frac{1}{2}} \right) \tag{6}$$

$$V_{TH0} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}} = V_{TH} \; when \; V_{SB} = 0 \tag{7}$$

Subthreshold (weak inversion):

When $V_{GS} \approx V_{TH}$, the device enters weak inversion, and the drain current follows:

$$I_D = I_0 \exp \frac{V_{GS}}{\xi V_T} \tag{8}$$

where I_0 is proportional to W/L, $\xi > 1$, and $V_T = kT/q$. The device is said to operate in *weak inversion*, as opposed to *strong inversion*.

Small-signal model:

Figure 2 shows the small-signal model of NMOS [1]. The corresponding small-signal model for a PMOS transistor can be derived in the same way.

Equations:

In triode region, a MOSFET behaves as a voltage-controlled resistor:

$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})} \tag{9}$$

In saturation, the transconductance is:

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \tag{10}$$

Equivalent expressions include:

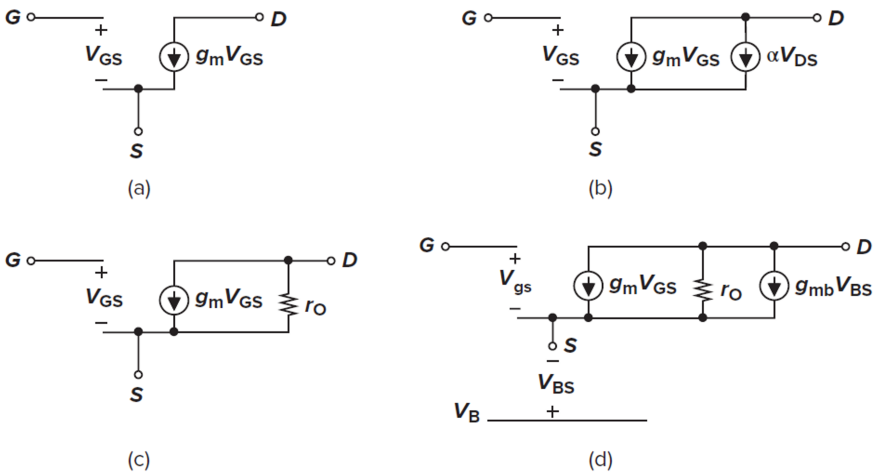


Figure 2: (a) Basic MOS small-signal model; (b) channel-length modulation represented by a dependent current source; (c) channel-length modulation represented by a resistor; (d) body effect represented by a dependent current source.

$$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH}) \tag{11}$$

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} \tag{12}$$

$$g_m = \frac{2I_D}{V_{GS} - V_{TH}} \tag{13}$$

The output resistance due to channel-length modulation is:

$$r_o = \frac{1}{g_{ds}} = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \times \lambda} \tag{14}$$

Given $\lambda V_{DS} << 1$, we have

$$r_o = \frac{1}{g_{ds}} = \frac{1 + \lambda V_{DS}}{\lambda I_D} \approx \frac{1}{\lambda I_D} \tag{15}$$

The body-effect transconductance is:

$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = gm \frac{\gamma}{2\sqrt{2\Phi_F + V_{SB}}} = \eta\ g_m \tag{16}$$

Simulation results:

Long-channel NMOS:

The circuit in Figure 3 is used to evaluate the small-signal, low frequency behavior of an NMOS device. Because capacitances are not considered in this section, DC-sweep analysis is employed.

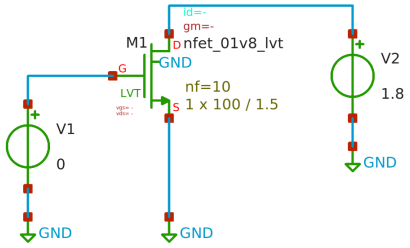
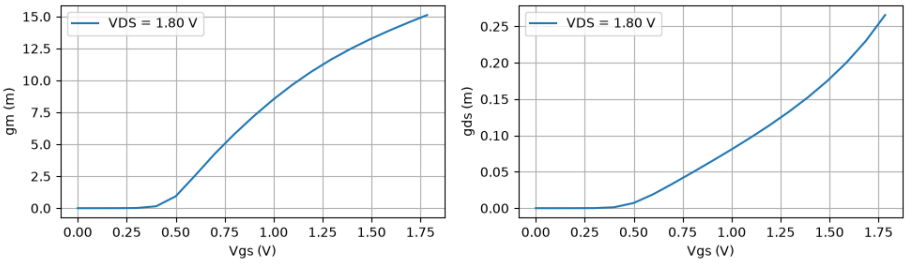


Figure 3: Simple NMOS circuit used for DC-sweep analysis

Here, W/L = 100u/1.5u, number of gates (fingers) are 10, and the VGS sweep ranges from 0 to 1.8V. Figure 4 shows gm and gds of the NMOS versus VGS.



(a) g_m versus V_{GS} (b) g_{ds} versus V_{GS}

Figure 4: Variation of g_m and g_{ds} versus V_{GS} for $V_{DS} = 1V$.

In this circuit, $V_{TH} \approx 470mV$. As V_{GS} increases, both g_m and g_{ds} increase as expected. It's interesting to see the gds – Id relationship as shown in Figure 5. The curve is nearly linear, and according to Equation 15, its slope corresponds to the channel-length modulation coefficient λ.

The intrinsic small-signal gain of a MOSFET is defined as

$$A_v = g_m r_o = \frac{g_m}{g_{ds}} \tag{17}$$

The intrinsic gain of a long-channel NMOS transistor is shown in Figure 6. As can be seen, in saturation region, the gain peaks at approximately $V_{GS} = 0.56V$.

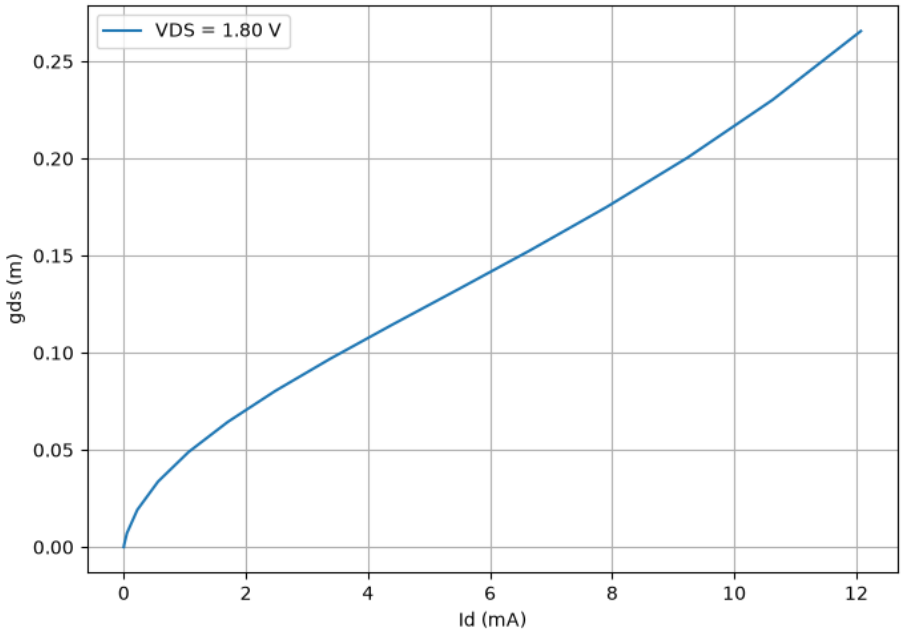


Figure 5: Variation of g_{ds} versus Drain current I_d for $V_{DS} = 1.8V$.

This corresponds to an overdrive voltage of $V_{ov} = V_{GS} - V_{TH} \approx 100mV$, at which the device achieves its maximum intrinsic gain. As V_{ov} increases further, the intrinsic gain decreases.

For V_{ov} below roughly 80mV the device operates in the *weak-inversion* region. When $V_{GS} < V_{TH}$, the transistor is off and the concept of intrinsic gain is no longer meaningful.

Short-channel NMOS:

For channel length below $1\mu m$ (submicron), MOSFET behavior is significantly affected by short-channel and other higher-order effects. Figure 7 illustrates how these effects alter the small-signal parameters of a device with $W/L = 10\mu m/0.15\mu m$. A detailed discussion of short-channel devices will be presented in a subsequent article.

[1] B. Razavi, *Design of analog CMOS integrated circuits*. Available: https://www.mheducation.com/highered/product/design-of-analog-cmos-integrated-circuits-razavi.html

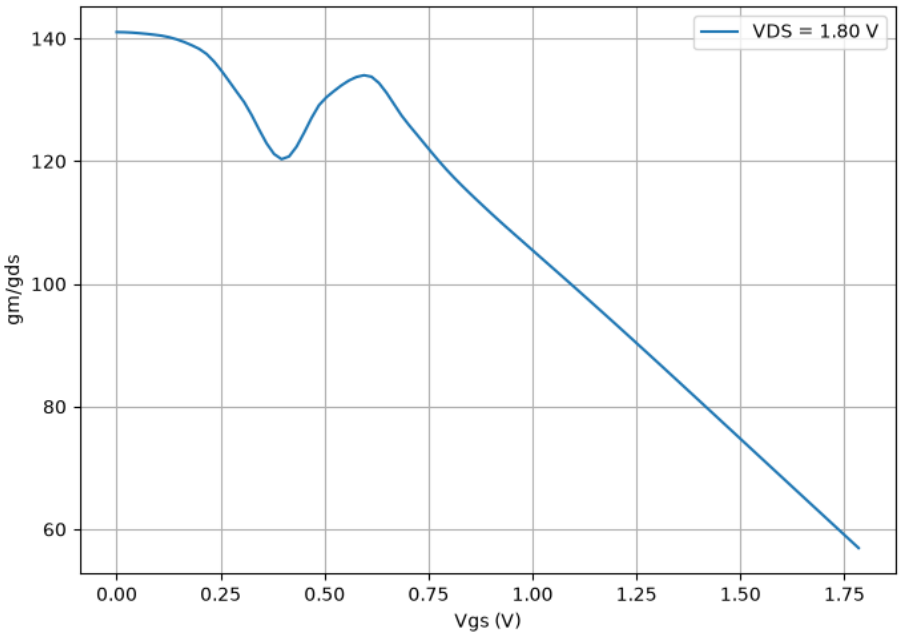
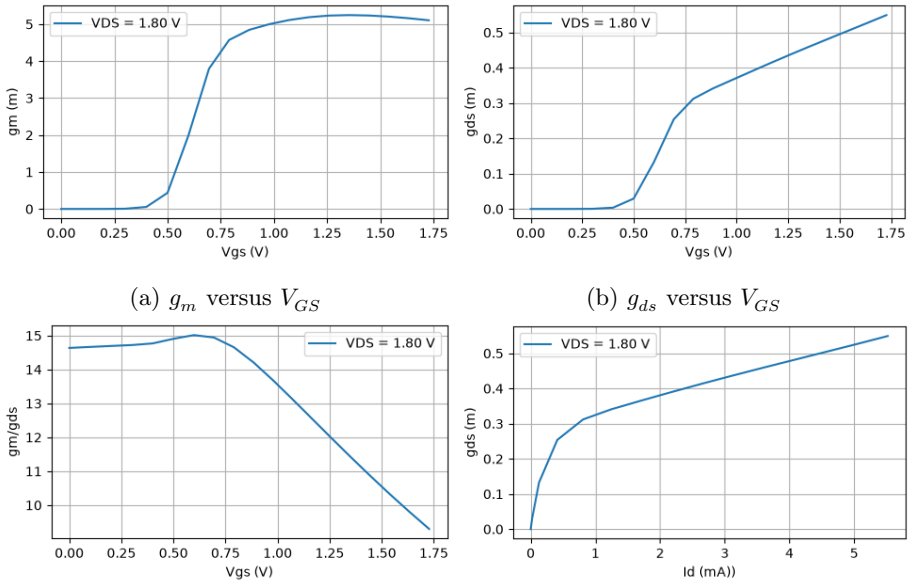


Figure 6: Intrinsic small-signal low-frequency gain of NMOS with W/L = 100u/1.5u.



(a) g_m versus V_{GS}

(b) g_{ds} versus V_{GS}

(c) g_m/g_{ds} versus V_{GS}

(d) g_{ds} versus I_d

Figure 7: Variation of small-signal parameters versus V_{GS} for short-channel NMOS where $W = 10\mu m$ and $L = 0.15\mu m$.